

Coding-Agent Deployment Systems

A Predeclared Cross-Harness Evaluation
with a Post-Hoc Luna Solo Supplement

A thesis-style technical report and corrective extension

Date: 15 August 2026
Host: Linux x86_64, 4 logical CPUs, 15.5 GiB RAM
Cursor Agent CLI: 2026.08.11-e8db854, authenticated subscription
OpenCode: 1.18.18, authenticated subscription
New outcome runs: 35 predeclared plus 10 post-hoc exploratory
Prior controlled runs: 40 in OpenCode and Pi

The complete new protocol, raw streams, hidden-grader outputs, hashes, and analysis are under
reproducibility/cross-harness/.

Abstract

An earlier experiment on this machine found that a GPT-5.6 Sol/xhigh parent with three bounded Terra/medium workers completed one separable coding workload faster than Sol/xhigh alone in OpenCode and Pi. That study omitted a strong external baseline: a fast single-agent system in Cursor CLI.

This extension predeclared seven deployment arms in a local protocol frozen before outcomes and ran five randomized complete blocks, producing 35 fresh outcome runs on the same hidden-graded, three-package workload. The arms were solo Cursor Grok 4.5 high, Grok 4.6 high, and Grok 4.6 xhigh; OpenCode Sol/xhigh solo; bounded Sol+Terra/medium; bounded Sol+Luna/max; and production-like Sol+Luna/max. Cursor’s Task tool was excluded, and raw streams confirmed zero subagent calls. The workload was deliberately separable, favoring parallel delegation.

The pre-outcome freeze was local rather than an externally archived immutable preregistration. Current analysis checks artifact and score integrity, but cannot prove that the source files were unchanged before execution.

Solo Grok 4.5 high was fastest and highest-scoring: 113/120 hidden checks in 109.5 seconds mean agent time. OpenCode Sol/xhigh solo scored 112/120 in 486.8 seconds. Grok 4.5 reduced aggregate agent time by 77.5%; every block favored Grok, and the mean paired reduction was 77.0% with a descriptive 95% small-sample interval of 69.2–84.7%. Bounded Terra scored 106/120 in 405.1 seconds. Bounded and natural Luna scored 110/120 in 949.4 and at least 1190.2 seconds mean; natural Luna timed out once. Grok 4.6 high and xhigh did not improve aggregate quality and were 4.0 and 5.7 times slower than Grok 4.5.

After the parent analysis, a requested post-hoc supplement randomized Luna/max solo and Luna/xhigh solo in five new blocks. Luna/xhigh scored 110/120 in 750.9 seconds mean agent time; Luna/max scored 109/120 in 779.4 seconds. Xhigh used 3.7% less aggregate time but was faster in only two blocks, and its paired interval crossed zero. Against non-concurrent historical outcomes, Grok used 85.4–86.0% less aggregate time and Sol used 35.2–37.5% less, with higher quality. This corroborates but does not alter the frozen parent selection.

No arm achieved a perfect run. The result is therefore not permission to skip verification. It is a deployment decision: default to solo Cursor/Grok 4.5 high for the tested workflow; use OpenCode Sol/xhigh solo if OpenCode is required; do not replace Sol with Luna based on the post-hoc solo screen; disable Luna/max subagents; and treat bounded Terra as experimental rather than a default.

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1 Decision

Operating recommendation. Default to solo Cursor Agent with Grok 4.5 high. Keep Task/subagent invocation disabled unless a task-specific, hidden-graded comparison demonstrates an advantage. If OpenCode is required, use Sol/xhigh solo with `task` denied. Disable Luna/max subagents. Treat Luna solo as unsupported by the post-hoc screen. Treat bounded Terra as an experimental low-risk mode, not a security-sensitive default.

This practical production command distills the tested default:

```
agent -p --force --trust \
  --model cursor-grok-4.5-high \
  --output-format stream-json \
  --exclude-tools task_tool_call \
  "<task>"
```

It is not the exact isolated invocation. The harness also used a fresh `CURSOR_CONFIG_DIR` and `--data-dir`, explicit workspace, `--single-turn`, `--sandbox disabled`, and flags disabling project config, indexing, and codebase references. Exact reproduction uses `run.mjs`.

The installed CLI accepted `--exclude-tools task_tool_call`, although that flag was absent from ordinary help output. Cursor documents automatic subagents, so model selection alone does not prove solo execution. The five outcome streams each contained one session ID and zero Task calls.

This policy is evidence-calibrated. It is strong for this fixture and observation window, moderate as a personal default, and not a universal model ranking. The post-hoc solo supplement also found historical Sol 35.2–37.5% faster than Luna/xhigh or Luna/max, with higher aggregate quality.

2 Why the Earlier Thesis Was Incomplete

The prior study asked whether subagents improved an OpenCode-centered workflow. Its within-harness findings were useful:

Prior harness	Single mean	Delegated mean	Time change	Quality
OpenCode 1.18.7, five pairs	19.5m	11.3m	−42.1% 108/120 vs 112/120 corrected	
Pi 0.80.6, three pairs	12.2m	8.0m	−33.9%	68/72 vs 66/72

Those results answer an architecture question conditional on using Sol and the tested harness. They do not answer whether to use OpenCode and GPT-5.6 at all. Sol/xhigh was treated as an established default rather than tested against Cursor/Grok. The previous report disclosed that limitation but still made an operating recommendation beyond its evidence.

This is not a cosmetic distinction. SWE-agent shows that agent-computer interface design can materially affect coding performance, while Agentless demonstrates that simple systems can outperform elaborate autonomous agents. The intervention relevant to a user is the complete deployment system: harness, model, tools, reasoning effort, and delegation policy.

3 Research Questions

- RQ1** Which tested solo Cursor/Grok configuration is quality-eligible and fastest on the frozen workload?
- RQ2** Does the best solo Cursor arm beat OpenCode Sol/xhigh alone on hidden quality and user-perceived completion time?
- RQ3** Does it beat the earlier recommended bounded Sol+Terra system?
- RQ4** Within current OpenCode, do Terra or Luna subagents improve time without unacceptable quality loss?
- RQ5** Do the tested higher-effort configurations improve quality enough to justify their latency?
- RQ6** What operating policy survives the combined old and new evidence?

After the parent study was complete, the user requested a separate exploratory question: how do Luna/max and Luna/xhigh perform as solo OpenCode owners? It is not retroactively designated a preregistered research question.

The estimand is deployable-system performance. This study does not identify a pure causal effect of concurrency or model family.

Capacity was asymmetric by design: OpenCode solo had 36 parent turns; bounded arms had 12 parent turns plus up to 36 child turns; natural Luna could continue until the wall cap. Cursor exposed no equivalent turn cap. These are usable architectures, not equal-compute treatments.

4 Preregistered Design

4.1 Experimental arms

Arm	Parent / solo	Children	Delegation policy
Cursor Grok 4.5	Grok 4.5 high	None	Task excluded
Cursor Grok 4.6 high	Grok 4.6 high	None	Task excluded
Cursor Grok 4.6 xhigh	Grok 4.6 xhigh	None	Task excluded
OpenCode solo	Sol/xhigh, 36 turns	None	Task denied
Bounded Terra	Sol/xhigh, 12 turns	3 Terra/medium	One package each; no recursion
Bounded Luna	Sol/xhigh, 12 turns	3 Luna/max	One package each; no recursion
Natural Luna	Sol/xhigh	Luna/max	Broad optional recursion

The natural arm reproduced the relevant former production configuration: only `general`, `explore`, and `scout`, all Luna/max, were available as subagents.

4.2 Workload and grading

Every run received a fresh copy of three independent JavaScript packages:

- bounded-concurrency mapping implementation;
- session and authentication security repair;
- event-pipeline comprehension producing eight exact facts.

Each package contributed eight hidden checks, 24 per run. Agents saw public tests. Graders were not copied into their workspaces, and exact grader paths did not appear in streams, but OS-level non-access was not enforced. The comprehension deliverable was fixed before outcomes at `comprehension/answer.json`; there was no score-improving path fallback.

The explicit package independence favors subagents. A solo winner therefore wins despite a workload designed to expose parallelism. The study says little about coupled refactors.

4.3 Sampling, randomization, and endpoints

Five randomized complete blocks were frozen before outcomes. Each contained all seven arms once. A seeded generator fixed the 35-run order; top-level runs were sequential. The prompt, fixture hash, arm set, model IDs, five-block sample, 30-minute cap, and decision rule were frozen before outcome inspection.

Primary quality was hidden checks passed. Primary performance was process wall time from agent launch to exit. Verified wall time added deterministic grading and produced nearly identical estimates. Checks were clustered within runs and were not treated as independent samples.

The predeclared operating rule required no timeout or abnormal exit, no model mismatch or forbidden final fixture change, aggregate quality within one check of the best arm, and at least 7/8 security checks in every run. Among eligible systems, lowest aggregate verified time won.

4.4 Pilot amendments

Two excluded pilots found harness defects before outcomes. Cursor rejected `--exclude-workspace-context` for this account, so it was removed for all Cursor arms. The first natural-Luna pilot could see benchmark-only worker definitions, so that arm was restricted to the user's three production agent names. Failed pilot artifacts are retained.

4.5 Post-hoc Luna solo supplement

On August 15, after the parent result was known, a separate manifest froze Luna/max solo and Luna/xhigh solo before their own outcomes. Both used OpenCode, a 36-step cap, denied Task, and received the same fixture, prompt, graders, and 1800-second timeout. Five seeded blocks randomized these two arms and produced ten sequential outcomes.

Max versus xhigh is randomized within the supplement. Comparisons to August 14 Grok and Sol are non-concurrent descriptive comparisons, not paired effects. The supplement does not change the parent manifest or operating-rule winner. Its 43 raw artifacts are hash-pinned; source/config hashes are an explicitly post-outcome snapshot, not proof of a pre-outcome source freeze.

5 Main Results

5.1 Arm-level outcomes

Arm	Checks	Mean	Median	Range	Timeouts
Grok 4.5 solo	113/120	109.5s	107.0s	89–140s	0
Grok 4.6 high solo	112/120	433.9s	427.4s	398–471s	0
Grok 4.6 xhigh solo	111/120	622.8s	592.9s	558–688s	0
Sol/xhigh solo	112/120	486.8s	468.3s	401–574s	0
Sol+Terra bounded	106/120	405.1s	402.0s	230–584s	0
Sol+Luna bounded	110/120	949.4s	880.9s	790–1346s	0
Sol+Luna natural	110/120	$\geq 1190.2s$	1083.9s	893 to $> 1800s$	1

Grok 4.5, Grok 4.6 high, and Sol solo passed the fixed gate. Grok 4.5 won mechanically on time. Grok 4.6 xhigh, Terra, and both Luna arms fell outside the one-check quality margin; natural Luna also timed out.

5.2 Grok 4.5 versus OpenCode Sol solo

Trial	Grok 4.5	Sol solo	Time saved	Scores
1	89.3s	553.4s	83.9%	22 vs 22
2	89.5s	401.0s	77.7%	23 vs 22
3	122.1s	468.3s	73.9%	23 vs 22
4	139.8s	437.6s	68.0%	22 vs 23
5	107.0s	573.6s	81.3%	23 vs 23

Aggregate agent-time reduction was 77.5%. Mean paired reduction was 77.0%, with descriptive 95% interval 69.2–84.7%. Mean absolute savings were 377.3 seconds, interval 275.0–479.5 seconds. Every block favored Grok. Aggregate quality was 113/120 versus 112/120: parity under the fixed margin, not evidence of general intelligence superiority. No speed-quality tradeoff was observed under the prespecified margin.

5.3 All preregistered contrasts

Block-level agent-time reductions for the candidate named first:

Contrast	T1	T2	T3	T4	T5
Grok 4.5 vs Sol	83.9%	77.7%	73.9%	68.0%	81.3%
Grok 4.5 vs Terra	81.9%	77.7%	61.5%	39.2%	81.7%
Terra vs Sol	11.1%	−0.2%	32.2%	47.5%	−1.8%
Terra vs Luna	44.7%	52.1%	59.9%	82.9%	33.7%
Sol vs natural Luna	50.2%	62.2%	47.6%	$\geq 75.7%$	47.1%

Contrast	Aggregate	Mean [95% interval]	Quality	Faster	Perfect	Timeouts
Grok 4.5 vs Sol	77.5%	77.0 [69.2,84.7]	113–112	5/5	0–0	0–0
Grok 4.5 vs Terra	73.0%	68.4 [45.6,91.2]	113–106	5/5	0–0	0–0
Terra vs Sol	16.8%	17.7 [−8.9,44.4]	106–112	3/5	0–0	0–0
Terra vs Luna	57.3%	54.7 [31.7,77.7]	106–110	5/5	0–0	0–0
Sol vs natural Luna	≥ 59.1%	56.6 [41.2,71.9]	112–110	5/5	0–0	0–1

Natural-Luna trial 4 is right-censored. Its reduction is a lower bound; the interval using the timeout cap is descriptive, not a completion-time interval.

Mean absolute agent-time savings with 95% intervals were: Grok 4.5 versus Sol, 377.3 seconds [275.0,479.5]; Grok 4.5 versus Terra, 295.5 [102.1,489.0]; Terra versus Sol, 81.7 [−36.8,200.2]; Terra versus Luna, 544.3 [138.7,949.8]; and Sol versus natural Luna, 703.4 [233.9,1173.0] using the capped timeout and thus not an exact completion-time interval.

5.4 Grok 4.5 versus bounded Terra

Grok was faster in all five blocks. Aggregate agent-time reduction was 73.0%; mean paired reduction was 68.4%, interval 45.6–91.2%. Quality was 113/120 versus 106/120. Terra scored only 5/8 on security in two runs and failed the safety gate. The missing external baseline therefore reverses the old operating recommendation.

5.5 Delegation within current OpenCode

Bounded Terra reduced aggregate agent time by 16.8% versus Sol solo but was faster in only three of five blocks. Mean paired reduction was 17.7%, interval −8.9–44.4%. Terra lost six checks, 106/120 versus 112/120. Excluding its transient-test-edit deviation yielded a similar clean-block estimate: 20.1% aggregate reduction, interval −12.6–57.1%, and 84/96 versus 90/96 checks.

Terra was 57.3% faster than bounded Luna but lost four checks. Excluding both bounded protocol-deviation blocks left three pairs: Terra remained 45.6% faster and scored 64/72 versus 66/72.

Sol solo was faster than natural Luna in every block. The capped intention-to- treat advantage was at least 59.1%; one natural run is right-censored at 1800 seconds. Across the four normally completed pairs, Sol remained 51.9% faster and scored 89/96 versus 88/96.

5.6 Higher-effort configurations did not dominate

Grok 4.5 reduced time by 74.8% versus Grok 4.6 high and 82.4% versus Grok 4.6 xhigh while scoring 113/120 versus 112/120 and 111/120. This changes model generation and effort, so it does not isolate a causal effort effect. The earlier same-model screen also found no gain from Terra/high over Terra/medium, while Luna/max was slow. Operationally, higher-effort tested configurations did not guarantee better quality.

5.7 Post-hoc Luna solo results

Exploratory arm	Checks	Mean	Median	Range	Timeouts
Luna/xhigh solo	110/120	750.9s	780.5s	621–816s	0
Luna/max solo	109/120	779.4s	768.3s	682–909s	0

Every run had exactly one observed Luna parent, no child session or Task call, a normal exit, and only allowed file changes. Xhigh used 3.7% less aggregate time and scored one check higher than max, but was faster in only two of five randomized blocks. Mean paired reduction was 3.1%, descriptive 95% interval -11.5 – 17.7 %; mean savings were 28.5 seconds, interval -86.2 – 143.2 . There is no clear latency effect between these effort settings.

Non-concurrent baseline	vs Luna/xhigh	vs Luna/max	Quality
Grok 4.5 solo	85.4% less time	86.0% less	113 vs 110 / 109
Sol/xhigh solo	35.2% less time	37.5% less	112 vs 110 / 109

These historical gaps reinforce the operating choice but are not randomized within-block effects because host and service time differed.

6 Quality and Resource Analysis

6.1 Systematic misses

No preregistered arm achieved 24/24, and all seven preregistered arms solved comprehension perfectly:

Arm	Implementation	Security	Comprehension
Grok 4.5	35/40	38/40	40/40
Grok 4.6 high	35/40	37/40	40/40
Grok 4.6 xhigh	36/40	35/40	40/40
Sol solo	37/40	35/40	40/40
Bounded Terra	35/40	31/40	40/40
Bounded Luna	35/40	35/40	40/40
Natural Luna	35/40	35/40	40/40

Every Grok 4.5 run missed full invalid-concurrency validation; two also missed a redirect parser edge. Every Sol run missed the redirect edge; three missed invalid concurrency. Terra additionally missed profile-patch protections in two runs. Public tests were not sufficient to certify completion. Contract-focused tests remain mandatory, especially around nullability, parser layers, authentication, redirects, cancellation, and cleanup races.

Neither exploratory Luna solo arm produced a perfect run. Both scored 35/40 on implementation and 35/40 on security, systematically missing invalid- concurrency and redirect checks. Luna/xhigh scored 40/40 on comprehension; Luna/max scored 39/40 after one `dedupeKey` miss.

6.2 Telemetry

Across five Cursor runs, Grok 4.5 reported 135,151 input, 51,494 output, and 649,600 cache-read tokens. Grok 4.6 high reported 523,892, 143,195, and 1,432,192; xhigh reported 889,517, 197,714, and 2,333,568. Cursor emitted no authoritative subscription charge or separate reasoning-token field.

OpenCode database telemetry and the prior nominal rate card gave:

Arm	Sessions	Input	Output	Reasoning	Proxy cost
Sol solo	5	212,008	37,775	79,682	\$5.117
Bounded Terra	20	368,169	67,375	64,343	\$4.720
Bounded Luna	20	691,841	97,013	303,225	\$5.741
Natural Luna	21	890,564	102,395	403,134	\$6.432
Luna/xhigh solo*	5	426,285	60,729	130,879	\$1.896
Luna/max solo*	5	373,067	49,511	153,038	\$1.891

The starred rows are from the post-hoc supplement. Their lower nominal proxy reflects the Luna rate card, not faster completion or an actual invoice.

These are API-equivalent proxies, not OAuth subscription invoices, and Cursor and OpenCode token semantics differ. Time, hidden quality, and rate-limit behavior are more actionable than invented cross-vendor dollars.

7 Protocol Deviations and Integrity

All 35 outcomes exist, match the frozen order, share the fixture and prompt digests, and match the stored grader outputs and current artifact hashes. A manifest verified 163 raw artifacts before reanalysis. Requested and observed models matched; Cursor made zero Task calls; final changes were in scope. No exact grader filename or grader directory appeared in a model stream.

Adversarial review found two bounded deviations:

- Terra trial 2 temporarily edited `security/public.test.js`. The parent resumed the same worker in a fourth Task call and restored it before grading.
- Luna trial 4 resumed an implementation worker after it reported its configured step cap, weakening a strict 12-turn interpretation.

Both remain in intention-to-treat estimates and are excluded in clean-block sensitivity estimates. Neither affects the Grok-versus-Sol contrast.

Isolation was policy-based, not OS-enforced. Workspaces were fresh and graders were outside them, but Cursor used `--sandbox disabled`, processes inherited the host environment, and harness credentials were present. No observed path leakage is not proof of non-access. Future replication should use containers or namespaces and broker credentials outside model-executed shells.

The post-hoc supplement retained ten outcomes plus streams, stderr, and grader outputs. Its 43-file manifest verified before reanalysis. Every observed parent was Luna with the requested xhigh or max variant; all runs were solo and in scope. Its analyzer also verifies the parent 163-file manifest and records the parent `runs.json` digest. Independent audit found no arithmetic or current artifact-integrity error; methodological limits remain material. Source/ config hashes were captured after

outcomes and are disclosed as such.

8 Reconciling Old and New OpenCode Evidence

Study	Sol mean	Terra mean	Reduction	Quality change
August 12, OC 1.18.7	19.5m	11.3m	42.1%	+4/120 corrected
August 14, OC 1.18.18	8.1m	6.8m	16.8%	−6/120

The difference is evidence, not an inconvenience. Provider sampling, server conditions, harness versions, and model behavior make agent-system effects temporally unstable. One fixture estimates repeatability in one observation window, not a permanent architecture constant.

The synthesis is narrower:

1. parallel delegation can compress critical paths for independent work;
2. the benefit can be modest or unreliable even on a favorable fixture;
3. worker quality and parent correction determine whether speed survives;
4. a faster solo harness/model can dominate the delegated architecture;
5. architecture selection needs external baselines and periodic retesting.

The August 15 solo supplement adds a same-harness descriptive check: Luna/xhigh and Luna/max averaged 12.5 and 13.0 minutes versus the August 14 Sol mean of 8.1 minutes. Sol was not rerun concurrently, so this corroborates rather than estimates a randomized Sol-versus-Luna effect.

9 Adversarial Defense

9.1 One synthetic fixture cannot select a universal winner

Sustained. This is a personal deployment decision. Multi-language, SWE-bench-scale, coupled, interactive, and long-horizon replication is needed for broad claims.

9.2 The workload was engineered for subagents

Sustained, but conservative for the result. Explicit independence makes delegation easier. It cannot explain the solo arm’s advantage.

9.3 Harness and model are confounded

Sustained by design. The user selects complete systems. A factorial study would be needed to attribute the advantage to Grok versus Cursor’s loop.

9.4 Cursor had no turn cap

Sustained. No equivalent was exposed. A common wall cap estimates which deployable system manages work efficiently.

9.5 Five blocks are too few

Sustained for population inference. The Grok–Sol gap was large and directionally consistent, but task-distribution coverage remains absent.

9.6 Scores differ by one check

Sustained. No intelligence superiority is claimed. Grok met quality parity while taking far less time. Every system still requires verification.

9.7 The timeout biases natural-Luna time

Sustained. It is right-censored. The capped result is a lower bound on Sol’s advantage; four completed pairs still favor Sol by 51.9%.

9.8 Subscription economics and isolation are incomplete

Sustained. Actual plan allowances should be monitored. A future study needs OS isolation and credential brokering. This report does not manufacture cross-vendor prices or claim proof of grader non-access.

9.9 Luna solo was added after the parent result

Sustained. The supplement is post-hoc relative to the parent decision and cannot repair the omission retroactively. Its two-arm order was frozen before those ten outcomes, so xhigh versus max is randomized within the supplement. Grok and Sol comparisons remain non-concurrent and descriptive.

10 Operating Policy

10.1 Default mode

Use solo Cursor/Grok 4.5 high. Keep Task disabled when latency and attribution matter. Give one coherent task, inspect the diff, and require tests.

10.2 Verification gate

1. Run repository tests rather than trusting the completion summary.
2. Add focused checks for contract boundaries omitted by public tests.
3. Inspect changed-file scope, including tests and configuration.
4. Review security parsing and cancellation paths independently.
5. Reject “done” when final edits were not followed by rerun checks.

10.3 OpenCode mode

Use Sol/xhigh solo with Task denied if OpenCode is needed. The post-hoc solo screen does not support replacing Sol with Luna/xhigh or Luna/max. Do not expose broad Luna agents by default. Preserve bounded Terra only as an explicit experiment for low-risk independent packages with deterministic tests and parent review; the current evidence does not support security-sensitive use.

10.4 Rebenchmark triggers

Repeat a frozen comparison after a major harness or model change, subscription routing change, primary-language change, repository-scale change, or a shift from independent packages to coupled refactors. Compare benchmark predictions against actual completion and correction logs.

11 Threats to Validity

11.1 Internal validity

Sampling was nondeterministic. Harness, model, and effort were bundled. Two bounded runs deviated. Five blocks cannot balance every time trend. Isolation was not OS-enforced. The Luna supplement was requested after the parent result, and its source hashes were captured post-outcome.

11.2 Construct validity

Hidden checks measure selected edges, not maintainability or complete security. Wall time omits interactive steering. Nominal OpenCode cost is not billing. Repeated fixture completion overrepresents deterministic tasks.

11.3 External validity

One JavaScript fixture, machine, account, and three-day window were studied. Separability favors delegation. No large monorepository, UI task, migration, dependency upgrade, or multi-hour task was tested. Cursor subagents were not tested because the counterclaim concerned Cursor solo.

11.4 Statistical conclusion validity

Five blocks are imprecise for modest effects. Checks are clustered within runs. Best-Cursor selection used a frozen quality-first rule but still evaluated three variants on one fixture. The natural timeout is censored. Luna xhigh versus max also has five blocks, and its historical comparisons are not paired.

12 Reproducibility

The private research archive contains the frozen manifest, protocol and pilot amendments, post-outcome deviation ledger, isolated OpenCode config, resumable runner, 35 consolidated records, per-run raw streams and graders, a 163-file SHA-256 manifest, and standard-library-only analysis. The

`exploratory/luna-solo/` supplement contains its separate frozen two-arm manifest, isolated config, ten outcomes, 43-file hash manifest, and analyzer.

```
cd path/to/reproducibility/cross-harness
node analyze.mjs
cd exploratory/luna-solo
node analyze.mjs
```

The analyzer verifies current manifest and fixture digests, recomputes package scores from stored grader output, and checks count and order, prompt digests, model identity, Cursor Task calls, grader exits, per-run/consolidated equality, exact grader path appearances, final file scope, and raw artifact hashes. These checks do not prove pre-outcome source immutability or OS-level non-access to graders.

13 Conclusion

The original paper was right about mechanism but wrong to stop at the OpenCode boundary. Bounded delegation can compress a critical path. That does not make subagents the best workflow or a good default.

In the missing head-to-head comparison, solo Cursor/Grok 4.5 high was 77.5% faster than OpenCode Sol/xhigh solo, 73.0% faster than bounded Terra, and more accurate in aggregate than both. Luna/max subagents were substantially slower in the tested delegated configurations. The requested follow-up also found Luna/xhigh and Luna/max solo slower and lower-scoring than historical Sol and Grok; xhigh did not clearly beat max in the randomized supplement. Higher-effort configurations did not improve the decision outcome. The concern about subagent overhead is supported, and switching the OpenCode solo owner from Sol to Luna is not supported either.

Final thesis. Optimize the complete coding system, not the number of agents. Start with the fastest quality-eligible solo agent. Add delegation only after a task-specific hidden-graded comparison shows its integration overhead and quality risk are worth paying. On this fixture and observation window, that means Cursor Agent with Grok 4.5 high, no subagents, and mandatory verification.

References

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